

Doctoral Thesis

**Multimodal quantitative analysis of
functional materials: STEM and DFT
EELS simulation**

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Multimodal quantitative analysis of functional materials: STEM and DFT EELS simulation

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To every hand that helped, every voice that encouraged, and every silence that gave me space
to grow ...

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Transmission Electron Microscopy (TEM)

Scanning Transmission Electron Microscopy (STEM)

High-Angle Annular Dark Field (HAADF)

Electron Energy Loss Spectroscopy (EELS)

Energy Loss Near Edge Structure (ELNES)

Density Functional Theory (DFT)

Multislice Simulation

STEM Image Simulation

Dielectric Function

Antisite Defects

ATOMAP

EELS Simulation

